

Homework # 8: Monte Carlo Integration

Due April 18. Quiz 7 on Monte Carlo methods is on April 18.

1. An insurance company models the number of accidents during one day by a Poisson distribution with parameter θ . To account for the possibility of a large number of traffic accidents occurring on the same day, it uses Pareto(2,5) prior distribution with density $\pi(\theta) = 50\theta^{-3}$, for $\theta \geq 5$, instead of a mathematically simpler conjugate Gamma distribution. As a result, the posterior distribution is

$$\pi(\theta \mid \mathbf{X}) \sim e^{-n\theta} \theta^{\sum X_i - 3}, \quad \theta \geq 5,$$

and this distribution is not Gamma because $\theta \geq 5$.

Suppose that $\sum X_i = 16$ accidents occurred during $n = 4$ days.

- (a) Use Monte Carlo methods to estimate the normalizing constant C that makes $\pi(\theta \mid \mathbf{X}) = Ce^{-n\theta} \theta^{\sum X_i - 3}$ a proper density on $[5, \infty)$.
 - (b) Then estimate the posterior probability $\mathbf{P}\{\theta \geq 7 \mid \mathbf{X}\}$.
2. What size of a Monte Carlo study (the number of generated variables) is needed
 - (a) to estimate a posterior probability with a margin of error ± 0.001 with probability 0.95, if it is known that this probability cannot exceed 0.1?
 - (b) to estimate a posterior probability with a margin of error ± 0.001 with probability 0.95, if no other information is available?